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MASTER THESIS No.

**DEEP LEARNING-BASED ANIMAL
RE-IDENTIFICATION FOR NON-INVASIVE
WILDLIFE MONITORING AND
CONSERVATION**

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**DUBOKO UČENJE ZA PONOVRNU
IDENTIFIKACIJU ŽIVOTINJA U
NEINVAZIVNOM PRAĆENJU DIVLJIH VRSTA I
OČUVANJU PRIRODE**

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The master thesis assignment text in English goes here.

Ovdje dolazi tekst zadatka diplomskog rada na hrvatskom jeziku.

Thanks for the tea...

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1 Introduction

The conservation and effective management of wildlife populations depend crucially on accurate methods for monitoring animal abundance and diversity. Traditional approaches, including manual identification, tagging are often invasive, costly, time-consuming and impractical at large scales or in remote environments. In recent years, significant progress has been achieved through the application of automated computer vision methods for wildlife monitoring, particularly though individual animal re-identification (Re-ID). Animal Re-ID, which refers to the task of identifying and matching individual animals across multiple observations solely on visual appearance, has emerged as a viable non-invasive method for ecological studies and conservation efforts.[1] Despite the advancements, current visual Re-ID systems still face substantial challenges, especially when applied to real-world scenarios characterized by significant appearance variations, environmental clutter, occlusions, and limited training data. Current methods typically require extensive manual labeling effort or species-specific model training, restricting their applicability in general ecological monitoring and conservation task so developing an efficient, flexible Re-ID framework capable of generalizing to virtually any animal species without requiring significant manual annotation or retraining represents an important and currently unmet need in conservation technology. The main aim of this thesis is just that, to develop a generalizable, robust animal Re-ID and population estimation system capable of performing reliably across diverse animal species with minimal manual intervention. Specifically, this thesis attempts to create a method suitable for real-world scenarios characterized by limited labeled data, eliminating the requirement for extensive manual labeling which can often be very difficult and error-prone cause it requires a certain expertise. To achieve these goals, the research was structured around two interlinked objectives:

1. **Evaluation and optimization of robust classification methodologies:** Before addressing the complex task of population estimation, it was essential to establish a strong foundation for animal Re-ID that generalizes effectively across species. Publicly available datasets, notably WildlifeReID[2], were leveraged for evaluation and testing. This phase involved extensive experimentation with robust feature extraction methods, including DISK[3] and LightGlue[4] combined with dimensionality reduction and aggregation approaches such as PCA, GMM and Fisher Vectors.
2. **Development of population size estimation method:** Following the establishment of a reliable and generalizable individual classification framework, the ultimate goal was population counting. Implementation of the Nested Importance Sampling (NIS)[5], a human-in-the-loop statistical framework leveraging pairwise similarity scores produced by the general-purpose Re-ID model. Critically, this method requires minimal human input and does not require additional training or extensive labeling for new animal species or datasets, thus significantly reducing manual labor and expertise requirements in practical scenarios. Several potential weak points were discovered in the paper itself which were attempted to address in this thesis.

To further validate the practicality and effectiveness of the developed methods in real-world conditions, original data was collected specifically for this thesis. Two trail cameras were deployed on Medvednica mountain for approximately five months, gathering data of local wildlife populations (mostly wild boars and deer). The final and crucial stage of this thesis involved applying the developed Re-ID and population estimation framework directly to this newly collected, unlabeled dataset.

2 Related Work

In this chapter, we'll review existing approaches to visual animal re-identification (Re-ID) with an emphasis on methods intended to generalize across species. The literature will be organized into thematic categories, from traditional computer vision techniques to modern deep learning methods, including hybrid methods that combine the two. We also discuss key challenges in the field, such as limited data, pose variation and open-set identification. We'll synthesize the findings from prior work and emphasize the gaps that this thesis is trying to address.

2.1 Traditional Computer Vision Approaches

Early animal Re-ID systems relied on hand-crafted visual feature and local descriptors. A prime example is HotSpotter[6], a SIFT-based[7] algorithm that identifies individuals by comparing local keypoints in images. HotSpotter uses viewpoint-invariant SIFT descriptors and scores matches by emphasizing the most distinctive keypoints (the titular "hot spots") on an animal's coat pattern. [8] compared SIFT, SURF and SuperPoint on identifying sunfish individuals and demonstrated that the number of matching keypoints between a pair of images is a strong indicator for the likelihood that they contain the same individual. These traditional local-feature methods do not require training on identity labels and are, by definition, species-agnostic, making them applicable to a wide variety of species. HotSpotter and similar algorithms have been successfully used on numerous species with distinctive markings, including zebras, giraffes, jaguars, ocelots and leopards.[9] The drawback, however, is that purely hand-crafted methods cannot fully exploit any available training data since their matching heuristic is fixed and may miss more subtle or global cues.

2.2 Species-Specific Trait-Based Re-Identification

Another line of research focuses on species-specific visual traits for re-identification. Instead of analyzing the entire image or generic local patches, these approaches isolate particular biometric features unique to the species at hand, for example the ear shapes of elephants, the outline of a shark’s dorsal fin, or the contour of a whale’s tail (fluke), and use them for re-identification. Both traditional image analysis and deep learning have been employed to recognize such traits. For example, algorithms such as CurvRank[10] and finFindR[11] compute and match the shapes of whale flukes or shark fins to identify individuals. These methods often don’t require individual identity labels for training, rather a detector or feature extractor for the trait can be trained using only that trait’s annotations, then match those features between images. The advantage of trait-specific methods is that they leverage highly distinctive features (when they are present) and can work with minimal ID labeling. However, a major limitation is that they are species-specific so each system is tailored for identifying only one species and usually it can’t be transferred to other species. This lack of generality is a significant barrier to wider use as it requires developing a new method for each studied species. This thesis is motivated by overcoming this limitation by developing a method, based on existing research, that does not depend on any single species’ unique trait.

2.3 Deep Learning-Based Methods

With rise of deep learning, a number of methods have been proposed that learn features directly from image data, without manual feature engineering. Modern deep learning approaches for animal re-ID typically employ CNNs (Convolutional Neural Networks) or related architectures to produce an embedding (feature vector) for each image, such that images of the same individual are close in feature space while those of different individuals are far apart. These can be trained either as classification (training a CNN to classify a fixed set of known individuals) or via metric learning (training the network to differentiate between same-individual vs. different individual image pairs). In general, metric learning approaches are favored for wildlife re-ID because they naturally handle open-set problems. A classification-based model treats re-ID as identifying one of N known individuals, which fails when a new individual outside the training set appears.

By contrast, metric-based methods learn a similarity function and can thus match new individuals by embedding comparison, without needing to retrain for new identities.[9]

2.4 Hybrid Approaches with Learned Local Features

A promising direction in animal re-id is to combine the strengths of traditional and deep learning methods into a hybrid pipeline. The idea is to use modern learned features from deep networks in place of the hand-crafted ones, but still employ them in a local matching and aggregation framework as in traditional approaches. A good example of this is the ALFRE-ID[9] pipeline (Aggregated Local Features for Re-Identification). ALFRE-ID uses a CNN extractor to extract local patch descriptors from an animal's coat pattern, then aggregates these local features into a global descriptor for the image.

2.5 Key Challenges and Limitations

Research in animal re-identification encounters several fundamental challenges that limit real-world performance. Key issues identified in the literature are:

Limited labeled data. Collecting individual-level labels is labor-intensive, so datasets often contain only a handful of images per subject, leading to overfitting and poor generalization.

Pose variation. Animals appear under diverse viewpoints and behaviors, drastically altering their visual features. Identification accuracy drops when profiles or unusual poses replace frontal views.

Occlusions. Vegetation, group interactions, or partial camera-trap captures often conceal distinguishing markings, hindering feature extraction.

Open-set identification. Field deployments must handle individuals unseen during training; closed-set models misclassify such cases. Open-set recognition methods highlight the need to reject unknown identities.

3 Datasets

Accurate and generalizable animal re-identification (Re-ID) system depends on the quality and diversity of datasets used for development and evaluation. This thesis evaluates the proposed approach on publicly available datasets, mostly from WildlifeReID-10k dataset collection[12]

3.1 WildlifeReID-10k

WildlifeReID-10k[12] is a benchmark dataset collection explicitly curated for evaluating animal Re-ID methods across a variety of species. The primary goal of this dataset is to enable systematic benchmarking of general-purpose animal Re-ID methods. It aggregates 37 existing datasets into a unified format to enable easy evaluation. In total, WildlifeReID contains 140,488 images of 10,772 unique individuals spanning 33 species. The methods described in this thesis were tested on 7 datasets from that collection: ATRW (tigers), CowDataset (cows), ElPephants (elephants), CTai (chimps), Chicks4FreeID (chickens), SealID (seals) and SeaStarReID2023 (sea stars). They were chosen because they provide the range of challenges of re-identification and the testing on all 37 datasets would be too computationally expensive.

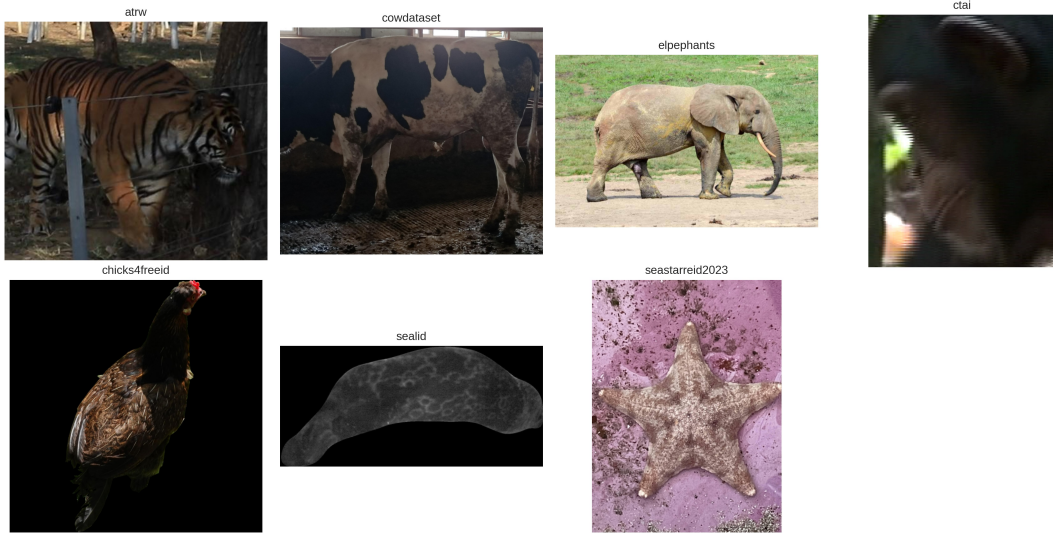


Figure 3.1: Example images from the datasets used in this thesis.

3.1.1 Train/Test Leakage Issues and Split Protocols

A central (and often not addressed) issue in animal Re-ID evaluation is train-to-test leakage caused by visually almost identical images (for example, consecutive video frames or images from camera traps taken in very quick succession during a single observation) ending up in both train and test sets. There are also mistakes in the datasets, so in some cases there are literal duplicates present. Because of this random splits often result in overly optimistic results that do not reflect real-world generalization.[12]

To address this, WildlifeReID-10k proposes two split strategies[12]:

- **Time-aware splits:** some of the datasets come with timestamps and this is used to split the images chronologically per identity using a cut-off date so that images from the same encounter do not leak across splits
- **Similarity-aware splits:** when timestamps are not available, an embedding model DINOv2[13] is used to cluster the images, leveraging the fact that near duplicates should lie close in the embedding space

3.1.2 MegaDescriptor Training Leakage

In this thesis I used the MegaDescriptor-L-384 (MD) model for extracting of global embeddings from images, this model was trained on a subset of datasets from WildlifeReID-

10k so for those datasets in evaluation I disregarded the time-aware/similarity-aware splits and used the exact same splits that the model was trained on to avoid additional data leaks. Those datasets are: ATRW, CTai and SealID and in later parts of the thesis will be marked with * to signify this. The Exception to this is the ELPephants[14] dataset which uses a random split.

3.1.3 Datasets Overview

Dataset	# images	# individuals	ratio	split type
ATRW*	5,415	182	29.8	Original MD split
CTai*	4,662	71	65.7	Original MD split
SealID*	2,080	57	36.5	Original MD split
CowDataset	1,485	13	114.2	Time-aware split
Chicks4FreeID	1,146	50	22.9	Similarity-aware split
SeaStarReID2023	2,187	95	23.0	Time-aware split
ELPephants	2078	276	7.5	Random split

Table 3.1: Datasets used for evaluation in this thesis.

3.1.4 Class balance visualizations

Another issue with Re-ID datasets is that they are typically highly imbalanced.



Figure 3.2: Class balance (number of images per individual) for the datasets used.

4 Methodology

4.1 Preprocessing

1. **Mantiuk Tone Mapping:** Lighting conditions in forests vary drastically, often resulting in overexposed or underexposed images. To mitigate this, **Mantiuk tone mapping** is applied to input images as suggested in [9]. This high-dynamic-range (HDR) to low-dynamic-range (LDR) mapping technique enhances local contrast and detail visibility, ensuring that key visual features (such as fur patterns) remain distinguishable even in bad lighting.
2. **Grounded SAM 2 Segmentation:** Traditional methods often rely on simple bounding box crops, which may include significant background clutter (trees, grass, rocks) that interferes with feature matching. To address this, a Grounded SAM 2 pipeline was implemented:
 - **GroundingDINO:** A text-to-bounding-box model is used to localize the specific animal within the frame using natural language prompts (e.g., “wild boar”).
 - **Segment Anything Model 2 (SAM 2):** Using the boxes from GroundingDINO as prompts, SAM 2 generates precise pixel-level segmentation masks for the animal.
3. **Background Removal with Soft Masking:** The computed masks are used to remove the background. Then, a soft masking technique is applied to the edges of the segmentation. This prevents the creation of harsh, artificial boundaries that could be misinterpreted as visual features by keypoint detectors.



Figure 4.1: Overview of the segmentation-based preprocessing pipeline used for background removal.

Some datasets from WildlifeReID-10k already came segmented so this pipeline was used only where judged as needed.

4.1.1 Mantiuk

bla bla

4.1.2 Segmentation

To control for background bias, a segmentation model has been employed to separate animals from their backgrounds. The resulting animal masks are fed into the feature-extraction pipeline alongside unsegmented originals. Comparing the results for both segmented and unsegmented methods allows us to quantify the impact of background removal on re-id performance and to probe for background bias.

4.2 Feature Extraction

Feature extraction is a critical component of animal re-identification pipelines, converting raw image data into structured descriptors that capture distinctive visual characteristics. In this chapter I will be examining the theoretical principles which guided the choice of a method for detecting and extracting local and global features from images.

This thesis uses two complementary feature families:

- **Global embeddings** (CNN descriptors computed from the entire image crop), used as a strong retrieval signal and as an input for fusion.

- **Local features** provide sets of keypoints and descriptors that capture fine-grained details

These two methods of feature extraction are evaluated both independently and in combination in later experiments.

4.2.1 Global Embeddings

Global embeddings represent an image as a single fixed-length vector produced by a pre-trained network. In the case of this thesis I used MegaDescriptor-L-384[2]. The pipeline that was used for the model is following:

1. load image crop (optionally after tone mapping / segmentation as described in Key Challenges and Limitations)
2. resize to the model's expected resolution (in this case 384×384 px)
3. convert to tensor and normalize by channel-wise mean/std
4. do a forward pass and compute the embeddings
5. save the embeddings in a .pkl format for later use

r

4.2.2 Local Feature Detection

Local features are small image patches, typically around corners, edges, or blobs that remain approximately stable under moderate changes in viewpoint, illumination and scale. Robust detection is critical in wildlife datasets, where pose, lighting and background vary dramatically from shot to shot. When we try to match two photos of the same animal, the first step is to pick out image patches that look the same even if the camera moved, zoomed, or the lighting changed. Those keypoints are what we call local features.

- **Corner detectors:** Harris corners [15] and the Shi-Tomasi update [16] look for pixels where the brightness changes strongly in two directions at once, basically

sharp corners. They are quick and accurate, but on their own they do not handle large zoom differences so we have to repeat the test at several image scales.

- **Blob detectors:** Instead of sharp points, blob detectors look for spots that are brighter or darker than their surroundings at many sizes. The Laplacian-of-Gaussian (LoG) scale space [17] does this exactly while the Difference-of-Gaussians (DoG) used in SIFT [7] is a fast approximation of it giving it scale invariance.
- **Learning-based detectors:** Recent methods use convolutional neural networks to learn both where to place keypoints and how to describe them. SuperPoint [18], R2D2 [19] and DISK [3] are three popular examples. By training on millions of image pairs, they learn to ignore lighting changes and background clutter, problems that often appear in camera-trap photos.

Choosing good keypoints boosts *recall* (more shared points between two images) and *precision* (fewer points that match the wrong place). That solid foundation lets later steps like feature aggregation and geometric verification work much better. In this paper, I opted to use learning-based detectors because they generally perform better than traditional ones like SIFT.

4.2.3 Descriptors

Once keypoints are detected, we compute descriptors that summarize the appearance of that local region. An effective descriptor should be discriminative (distinguishing one region from another) and robust (invariant to moderate geometric transformations).

- **Gradient-based descriptors:** Classic methods such as Scale-Invariant Feature Transform (SIFT) or Histogram of Oriented Gradients (HOG) capture local gradient orientations and magnitudes. They perform well in textured areas like stripes or distinctive fur patterns.
- **Learned descriptors:** Deep learning approaches train neural networks to produce feature vectors from image patches. HardNet [20], for example, is trained on many image pairs so that descriptors of matching patches are close together while mismatches are far apart, encoding subtle visual cues that hand-crafted methods

might miss.

- **Affine normalization:** Methods such as AffNet estimate the local affine transform (stretch and skew) before computing the descriptor, so viewpoint changes affect the patch less.

In this thesis, I used two different methods for feature extraction, DISK and KeyNet+AffNet+HardN

4.2.4 DISK: Learning Local Features With Policy Gradient

DISK (DIScrete Keypoints) is an end-to-end trainable local feature framework that integrates detection and description into a single model. Unlike traditional pipelines with separate components, DISK uses a unified architecture trained with reinforcement learning to maximize correct feature matches. It outputs keypoint heatmap and a 128-D descriptor vector.

DISK operates by simultaneously generating:

- **Keypoint Heatmaps:** Dense probability maps indicating optimal keypoint locations.
- **128-dimensional Descriptors:** Robust, discriminative vectors characterizing local appearance.

The unified training ensures features are optimized specifically for reliable and accurate matching, even under challenging environmental conditions commonly observed in wildlife photography.

4.2.5 KeyNet, AffNet, and HardNet

The second approach employs a combination of three specialized deep neural networks, each addressing distinct aspects of robust feature extraction and matching:

KeyNet

KeyNet[21] focuses on identifying stable and repeatable keypoints. It utilizes convolutional neural network layers trained explicitly to detect points in images that are con-

sistently recognizable under diverse transformations and viewing conditions. KeyNet significantly enhances keypoint stability, critical for reliable matching.

AffNet

AffNet complements KeyNet by predicting affine transformations to normalize identified keypoints. By correcting perspective distortions and other affine-related variations, AffNet ensures that the extracted descriptors are viewpoint-invariant, thereby significantly improving matching accuracy across diverse animal poses and camera angles.

HardNet

HardNet computes robust descriptors optimized for matching tasks. Unlike traditional descriptor approaches (e.g., SIFT or SURF), HardNet employs a metric learning strategy that ensures descriptors corresponding to the same keypoint in different images are close in descriptor space, while descriptors from different keypoints are distinctively separated. This contrastive learning approach makes HardNet descriptors highly discriminative.

The integrated KeyNet+AffNet+HardNet extraction process involves:

1. **Image Preprocessing:** Images are converted to grayscale and standardized in size to ensure compatibility with the neural network inputs.
2. **Keypoint Detection and Affine Normalization:** KeyNet identifies keypoints, followed by AffNet's affine transformation predictions to adjust keypoint localizations for geometric consistency.
3. **Descriptor Computation:** Adjusted keypoints are passed to HardNet, which computes robust, contrastively-learned descriptors.
4. **Storage:** Resulting descriptors and keypoints are separately saved in structured HDF5 files, preserving clear and efficient data organization.

4.2.6 Aliked

Aliked is also here and works good!

4.2.7 Storage Format and Management

Descriptors and keypoints are stored using the HDF5 file format, a hierarchical, binary storage format designed for managing large volumes of data. HDF5 ensures rapid read-write access, efficient compression to reduce storage demands, and structured data management.

In summary, feature extraction methods discussed in this chapter are the foundation for reliable and robust animal re-identification in this thesis, ensuring that subsequent analysis and matching procedures benefit from highly descriptive and discriminative image representations.

4.3 Feature Aggregation and Representation

This section describes the process of aggregating local feature descriptors extracted from animal images into compact global representations suitable for efficient comparison and re-identification. As described in [9] Fisher Vectors (FV) were utilized, built on PCA and Gaussian Mixture Models (GMMs).

4.3.1 Training PCA and GMM models

Feature descriptors extracted from images can be high-dimensional and noisy. Thus, dimensionality reduction and normalization are performed to improve efficiency and accuracy.

Initially PCA is applied to decorrelate and reduce the dimensionality of the descriptors. PCA projects the original descriptors onto a lower-dimensional subspace defined by the principal eigenvectors of the covariance matrix computed from the descriptors. This is also important later on for the computation of fisher vectors because they produce very large descriptors.[9]

After dimensionality reduction, a Gaussian Mixture Model (GMM) is trained on the reduced descriptors. A GMM models the descriptor distribution as a weighted sum of K Gaussian components, each parameterized by mean vectors $\{\mu_k\}$, covariance matrices

$\{\Sigma_k\}$, and weights $\{\pi_k\}$. Specifically, a GMM is defined as:

$$p(x) = \sum_{k=1}^K \pi_k \mathcal{N}(x|\mu_k, \Sigma_k), \quad (4.1)$$

where $\mathcal{N}(x|\mu_k, \Sigma_k)$ denotes the probability density function of the k -th Gaussian component. This codebook (the trained GMM) captures the visual vocabulary and statistical distribution of the reduced descriptors from the training images.[9]

In simpler terms, PCA finds the directions in which the descriptor values vary the most and keeps only those, so each feature becomes a shorter, uncorrelated vector that is faster to store and compare. Then we model how these compact vectors are distributed. GMM model assumes that the whole set of vectors can be grouped into a number of smooth clusters, each cluster described by its mean, spread and relative weight in the mixture. Together, these clusters form a codebook meaning that every new descriptor can be represented as a weighted combination of the nearest clusters, giving us both a compressed encoding and a way to compute similarity for matching or classification which is of crucial importance in re-identification task.

4.3.2 Fisher Vector Computation

Fisher vectors (FV) are an image representation technique that captures statistical deviations of local features from a visual vocabulary (codebook) modeled by GMM. FV measures how features differ from typical patterns in a reference GMM and they encode both first-order differences (mean shifts) and second-order differences (variance changes) in feature distribution. The output is a high-dimensional vector that acts as a "signature" of an image's unique feature distribution. Basically, given a set of descriptors extracted from a single image, FV encodes how these descriptors deviate from the trained GMM parameters. Fisher Vectors represent the gradients of the log-likelihood function with respect to the GMM parameters. Formally, given an image with descriptors $X = \{x_t\}_{t=1}^T$ where λ is a vector of GMM parameters and u_λ is a probability density function modeling the distribution of the data.

This gradient is formally known as the score function[9]. It measures the sensitivity of the log-likelihood to changes in the model parameters, essentially indicating how to

adjust the GMM to better fit the observed features.

$$G_{\lambda}^X = L_{\lambda} \log u_{\lambda}(X), \quad (4.2)$$

When constructing a FV for an image, a set of local features is assumed to be independent which means that the final descriptor can be constructed as a sum of Fisher Vectors for each local feature:[9]

$$G_{\lambda}^X = \sum_{t=1}^T L_{\lambda} \nabla_{\lambda} \log u_{\lambda}(x_t) \quad (4.3)$$

The matrix L_{λ} is derived from the Fisher Information Matrix and is used to "whiten" or to normalize the gradients[22], making them more discriminative and suitable for use in classification later on.

Following the standard practice, this vector is computed separately for means and variances, resulting in a final vector of dimension $2KD$, where K is the number of Gaussian components and D the dimensionality of the PCA-reduced descriptors. L2 and power normalization have also been applied since they have been shown[23] to generally improve the performance of the method.

FV have also been chosen because they have been shown to perform better than bag-of-words approaches.[24]

4.3.3 Parameter Choices and Rationale

Several key parameters influenced the effectiveness of Fisher Vectors for the re-identification task during the development:

- **Number of PCA components (N_{PCA}):** A balance between dimensionality reduction and retention of discriminative information.
- **Number of GMM components (N_{GMM}):** Controls the granularity of the visual codebook. More components typically yield richer representations but increase computational complexity.

- **Maximum descriptors sampled (*MAX_GMM_DESCRIPTOR*):** Limits the number of descriptors sampled for training PCA and GMM to manage memory usage and computational load.

These parameters were empirically determined based on experimenting and computational constraints. Specifically, the PCA dimension was selected to retain approximately 95% of descriptor variance. The number of GMM components was chosen to balance accuracy and computational efficiency. The best results were achieved, with 96 PCA components, 256 GMM components and 200,000 descriptors. The larger number of maximal components could perhaps prove better results but computational constraints were a limiting factor.

4.4 Geometric Verification

Geometric verification is used to make sure that matched keypoints between images truly correspond to the same physical locations on an animal reducing false positives that arise from visual similarities alone. This step is crucial in refining the initial matches obtained through descriptor-based feature matching by enforcing spatial consistency.

4.4.1 Theoretical Foundation

Local feature descriptors such as DISK[3] or KeyNet combined with AffNet and HardNet [21, 25, 20], offer robust initial matches based on visual similarity. However, matches that are generated purely from descriptor similarity may include mistakes, especially in visually cluttered or repetitive environments common in wildlife images. Geometric verification, as suggested in [9], addresses this by validating matches through spatial coherence checks.

4.4.2 Establishing Initial Correspondences

Initial correspondences are established through descriptor matching using LightGlue[4], an attention-based method. Initially methods like Lowe's ratio test were considered but matching with LightGlue proved significantly faster and more accurate, effectively eliminating the need for traditional filtering methods.

4.4.3 Transformation Estimation with RANSAC

The core of geometric verification involves estimating a homography transformation between matched keypoints using RANSAC (Random Sample Consensus). RANSAC iteratively samples minimal subsets of correspondences, computes candidate transformations and evaluates their agreement with the remaining matches. Matches that fall within a defined spatial threshold are considered inliers.

4.4.4 Scoring and Verification

Matches are assessed primarily by their inlier count, the number of matches consistent with the estimated homography, and their spatial distribution:

- **Inlier count:** Number of correspondences that agree with the estimated transformation; higher counts suggest more reliable matches.
- **Inlier ration:** Normalizes the inlier count by the total number of tentative matches,

$$r = \frac{|I|}{|C|},$$

where $|I|$ denotes the inlier count and $|C|$ the total correspondences.

- **Combined scoring:** Integrates geometric consistency with descriptor similarity (fisher vector distances) to leverage both spatial and appearance information.

Penalties for low inlier counts or poor geometric coherence discourage accepting borderline matches, improving the overall system reliability and accuracy.

5 Classification

Here we will describe in detail the process of classification

6 Population Size Estimation

Here I will, in detail, describe the process of population counting.

6.1 Introduction

While individual animal re-identification provides the foundation for wildlife monitoring, the ultimate goal in many conservation and ecological applications is estimating population sizes. Traditional clustering approaches for population estimation often produce inaccurate results when applied to re-identification systems, particularly when the underlying pairwise similarities are noisy or unreliable [5].

In this chapter I present the implementation and adaptation of Nested Importance Sampling (NIS) for population size estimation, originally proposed by Perez et al. [5]. The method provides a statistically principled approach to estimate the number of unique individuals in a dataset by leveraging pairwise similarity scores from the re-identification system developed in previous chapters. Critically, this work extends the original framework by implementing a fully automated variant that eliminates the need for human annotation during the estimation process.

6.2 Theoretical Foundation

6.2.1 Population Size as Connected Components

The population estimation problem can be formulated as counting connected components in a graph where vertices represent images and edges exist between images of the same individual. Formally, given a graph $G = (V, E)$ where vertices correspond to images and edges represent same-individual relationships, the number of clusters K equals the number of connected components:

$$K = \text{CC}(G) = \sum_{u=1}^n \frac{1}{1 + d(u)} \quad (6.1)$$

where $d(u)$ is the degree of vertex u (i.e., the number of edges connected to u) [5].

This elegant formulation arises from the observation that the graph G consists of a collection of cliques, where each clique represents images of the same individual. The clique containing vertex u has size $1 + d(u)$, and the total contribution of all vertices in this clique is $(1 + d(u)) \times \frac{1}{1+d(u)} = 1$. Thus, each clique contributes exactly 1 to the sum, yielding the total number of connected components.

This formulation provides the foundation for statistical estimation: instead of computing all pairwise similarities explicitly, we can sample vertices and estimate their degrees, then use importance sampling to obtain an unbiased estimate of the population size.

6.2.2 Nested Importance Sampling Framework

The NIS framework operates through a two-stage sampling process designed to efficiently estimate the population size while minimizing the number of pairwise comparisons required:

1. **Outer sampling:** Sample N vertices from the graph according to a proposal distribution $Q(u) \propto \frac{1}{1+\hat{d}(u)}$, where $\hat{d}(u)$ is the estimated degree based on approximate pairwise similarities.
2. **Inner sampling:** For each sampled vertex u_i , sample M neighbors according to $q_{u_i}(v) \propto \hat{s}(u_i, v)$, where $\hat{s}(u, v)$ is the approximate similarity between images u and v .

The population estimate is then computed as:

$$\widehat{\text{CC}}_{\text{NIS}}(G) = \frac{1}{N} \sum_{i=1}^N \frac{1}{Q(u_i)} \times \frac{1}{1 + \hat{d}(u_i)} \quad (6.2)$$

where $\hat{d}(u_i)$ is estimated through the inner sampling process [?].

The key insight behind this approach is that the proposal distribution $Q(u)$ biases sampling towards isolated vertices (those with low degrees), as these contribute most significantly to the cluster count. Similarly, the inner sampling distribution biases towards sampling nodes that are likely to be connected, maximizing the information gained from each pairwise query.

6.3 Automated Adaptation with Geometric Verification

6.3.1 Motivation for Automation

The original NIS framework relies on human annotation to verify whether pairs of images contain the same individual [5]. While this approach provides high accuracy, it presents practical limitations for large-scale deployments where manual annotation becomes prohibitively expensive or where expert annotators are unavailable. To address these limitations, this work develops a fully automated variant that leverages geometric verification as described in Chapter ??.

6.3.2 Similarity Computation

The approximate pairwise similarities $\hat{s}(u, v)$ are derived from the fused Fisher vector representations computed in the feature aggregation stage (Chapter ??). Cosine similarity is computed between normalized feature vectors and converted to a probability distribution using softmax normalization with temperature parameter $\tau = 0.5$:

$$\hat{s}(u, v) = \frac{e^{\hat{c}(u, v)/\tau}}{\sum_v e^{\hat{c}(u, v)/\tau}} \quad (6.3)$$

where $\hat{c}(u, v) = \frac{\Phi(x_u) \cdot \Phi(x_v)}{\|\Phi(x_u)\|_2 \|\Phi(x_v)\|_2}$ represents the normalized cosine similarity between feature vectors.

6.3.3 Automated Verification Process

Instead of relying on human verification of pairwise similarities, the automated system employs a gated verification process that combines Fisher vector distances with geometric verification. The process operates as follows:

1. **Initial filtering:** For each sampled pair (u, v) , verify that sufficient keypoints and descriptors are available for geometric verification.
2. **Geometric verification:** Apply the geometric verification pipeline (Chapter ??) to compute geometric distance and inlier count.
3. **Confidence assessment:** Combine geometric quality metrics with Fisher vector similarity to produce an overall confidence score:

$$\text{confidence} = \min\left(1.0, \frac{n_{\text{inliers}}}{20.0}\right) \quad (6.4)$$

$$\text{geometric_quality} = 1.0 - d_{\text{geometric}} \quad (6.5)$$

$$\text{overall_confidence} = 0.6 \times \text{geometric_quality} + 0.4 \times \text{confidence} \quad (6.6)$$

4. **Binary decision:** Accept the pair as a match if the overall confidence exceeds 0.5 and the number of inliers meets the minimum threshold.

This automated approach eliminates the need for human annotation while maintaining the statistical framework of the original NIS method.

6.4 Advantages and Limitations

6.4.1 Advantages

The automated NIS approach offers several key advantages over traditional clustering methods:

1. **Statistical guarantees:** Unlike heuristic clustering approaches, the NIS framework provides asymptotically unbiased population estimates with confidence intervals, enabling principled uncertainty quantification.
2. **Minimal supervision:** The automated mode eliminates the need for human annotation during population estimation, making it practical for large-scale deployments where expert annotation is unavailable or prohibitively expensive.

3. **Robustness:** The method handles noisy pairwise similarities better than traditional clustering approaches by explicitly modeling uncertainty through the sampling framework.
4. **Scalability:** Computational complexity scales as $N \times M$ rather than n^2 , where N and M are sampling parameters and n is the total number of images, enabling application to large datasets.

6.4.2 Limitations and Bias Considerations

The automated approach introduces several sources of bias that must be acknowledged:

1. **Geometric verification bias:** The reliance on geometric verification may systematically exclude certain types of valid matches, particularly:
 - Images with significant pose variations where keypoint correspondences are sparse
 - Low-texture regions with insufficient keypoints for reliable geometric estimation
 - Cases where the homography model is inappropriate for the underlying transformation
2. **Feature quality dependence:** The method's accuracy is fundamentally limited by the quality of the underlying Fisher vector representations and local feature extraction. Poor feature extraction can lead to both false positives and false negatives in the verification process.
3. **Threshold sensitivity:** The automated matching decisions depend on carefully tuned thresholds (geometric distance threshold, minimum inlier count, confidence threshold) that may not generalize across different species or imaging conditions.
4. **Systematic underestimation:** By requiring geometric consistency, the automated mode may systematically underestimate population sizes when valid matches are rejected due to overly restrictive geometric constraints. This represents a conservative bias that prioritizes precision over recall.

6.4.3 Bias-Variance Trade-off

The automated approach represents a fundamental trade-off between bias and variance. While the original human-in-the-loop method provides nearly unbiased estimates (assuming perfect human annotation), it requires substantial manual effort. The automated variant introduces systematic bias through the geometric verification gate but enables deployment at scale without human intervention.

This trade-off is particularly relevant in conservation applications where obtaining population estimates—even if slightly biased—may be more valuable than having no estimates at all due to the prohibitive cost of manual annotation.

6.5 Parameter Configuration

The implementation includes several key parameters that influence the bias-variance trade-off:

- N (**n_vertices**): Number of vertices sampled in the outer loop (default: 100)
- M (**n_neighbors**): Number of neighbors sampled per vertex (default: 10)
- θ (**gv_threshold**): Distance threshold for geometric verification acceptance (default: 0.75)
- $I_{\min}$ (**min_inliers**): Minimum number of RANSAC inliers required for acceptance

These parameters represent a trade-off between computational efficiency and estimation accuracy. Higher sampling rates (N , M) generally produce more reliable estimates at increased computational cost, while more restrictive geometric thresholds (θ , $I_{\min}$) reduce false positive matches but may increase systematic underestimation bias.

6.6 Integration with the Re-ID Pipeline

The population estimation module integrates seamlessly with the re-identification pipeline developed in previous chapters:

1. **Input:** Fused feature representations from the feature aggregation stage (Chapter ??)
2. **Optional inputs:** Keypoints and descriptors for geometric verification (Chapter ??)
3. **Processing:** NIS-based sampling and automated verification
4. **Output:** Population estimate with standard error and detailed verification statistics

This integration allows the system to provide both individual re-identification capabilities and population-level statistics from the same underlying feature representations, maximizing the utility of the computational pipeline.

6.7 Summary

This chapter presented the implementation of automated population size estimation using Nested Importance Sampling. The approach extends the original human-in-the-loop framework [5] by replacing human annotation with geometric verification, enabling fully automated deployment while maintaining the statistical rigor of the importance sampling framework.

While the automated approach introduces systematic bias through the geometric verification gate, it provides a practical solution for large-scale population estimation where human annotation is unavailable or prohibitively expensive. The method's effectiveness is evaluated experimentally in the following chapters, with particular attention to quantifying the trade-offs between automation and accuracy across different animal species and imaging conditions.

7 Experimental Setup

- Hardware
- Evaluacijske metrike
- Evaluacijske metrike za prebrojavanje individualaca

8 Results

- Rezultati klasifikacije (top-1, top-n acc), rezultati prebrojavanja individualnih životinja, rezultati prebrojavanja na mom datasetu
- Vizualizacije, tablice
- Usporedba utjecaja hiperparametara testiranih

9 Discussion

- Interpretation of results.
- Successes and limitations.
- Comparison with related work.

10 Conclusion and Future Work

- Summary of contributions and findings.
- Potential extensions and open questions.

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Abstract

Deep learning-based animal re-identification for non-invasive wildlife monitoring and conservation

Matej Marić

Reliable methods for monitoring wildlife populations are essential for effective conservation and ecological management. This thesis addresses the challenge of developing an automated visual system capable of identifying individual animals and estimating their population size across various species, particularly when limited labeled data is available. The primary goal was to establish a practical approach that can generalize effectively without the need for extensive manual annotation or species-specific model retraining. The developed methods were initially validated on publicly available benchmark datasets and subsequently tested using original data collected through trail cameras placed on Medvednica mountain in Zagreb, Croatia. The research specifically aimed at evaluating the system's practical applicability and accuracy in real-world conditions, focusing on local populations of wild boars (*Sus scrofa*) and roe deer (*Capreolus capreolus*).

Keywords: Animal Re-identification; Wildlife Monitoring; Population Estimation; Computer Vision; Conservation

Sažetak

Duboko učenje za ponovnu identifikaciju životinja u neinvazivnom praćenju divljih vrsta i očuvanju prirode

Matej Marić

Pouz dane metode praćenja populacija divljih životinja ključne su za učinkovito očuvanje i ekološko upravljanje. Ovaj rad bavi se izazovom razvoja automatiziranog vizualnog sustava sposobnog za identificiranje pojedinačnih životinja i procjenu veličine njihove populacije među različitim vrstama, posebno kada je dostupa ograničen broj označenih podataka. Primarni cilj bio je uspostaviti praktičan pristup koji se može učinkovito generalizirati bez potrebe za opsežnim ručnim označavanjem ili ponovnim treniranjem modela specifičnih za vrstu. Razvijene metode u početku su validirane na javno dostupnim skupovima podataka, a potom testirane korištenjem originalnih podataka prikupljenih putem kamera na stazama postavljenih na Medvednici u Zagrebu, Hrvatska. Istraživanje je posebno bilo usmjereno na procjenu praktične primjenjivosti i točnosti sustava u stvarnim uvjetima, s naglaskom na lokalne populacije divljih svinja (*Sus scrofa*) i srna (*Capreolus capreolus*).

Ključne riječi: Re-identifikacija životinja; Nadzor Divljači; Procjena Populacije; Računalni Vid; Očuvanje Prirode